Logistics System Database Design

This document outlines the database schema for the Centralized Cloud Platform. It is designed to support multi-tenancy (warehouses), Role-Based Access Control (RBAC), real-time telematics, and complex order lifecycles including "House Shifts" with labor and packing materials.

1. Core Organizational Tables

warehouses

- id (UUID, Primary Key)
- name (String)
- location (Point/Geography: Lat, Long)
- address (Text)
- operating_hours (JSON: {open, close})
- capacity_limit (Integer)
- created_at (Timestamp)

users (RBAC)

- id (UUID, Primary Key)
- warehouse_id (UUID, Foreign Key) - *Links staff to specific hubs.*
- role (Enum: MANAGER, WAREHOUSE_MANAGER, DISPATCHER, DRIVER, LABORER, CUSTOMER)
- full_name (String)
- email (String, Unique)
- password_hash (String)
- phone_number (String)
- is_active (Boolean)

2. Inventory & Resources

inventory_items (SKUs)

- id (UUID, Primary Key)
- warehouse_id (UUID, Foreign Key)
- sku_code (String, Unique)
- name (String)
- dimensions (JSON: {l, w, h, weight})
- quantity_on_hand (Integer)
- safety_threshold (Integer)
- location_code (String: e.g., "A-12-B")
- category (Enum: PRODUCT, UTENSIL, PACKING_MATERIAL)

vehicles

- id (UUID, Primary Key)
- warehouse_id (UUID, Foreign Key)
- license_plate (String)
- type (Enum: VAN, SMALL_TRUCK, LARGE_TRUCK)
- seat_capacity (Integer) - *Crucial for Laborer allocation.*
- load_capacity_kg (Float)
- status (Enum: MOVING, IDLE, MAINTENANCE)
- last_gps_lat (Float)
- last_gps_long (Float)

3. Order & Workflow Management

orders

- id (UUID, Primary Key)
- requester_id (UUID, Foreign Key)
- warehouse_id (UUID, Foreign Key)
- order_type (Enum: COMMERCIAL, PERSONAL_SHIFT)
- status (Enum: PENDING, APPROVED, PICKING, READY, DISPATCHED, COMPLETED, CANCELLED)
- pickup_address (Text)
- destination_address (Text)
- geofence_polygon (JSON/Geography)
- total_cost (Decimal)
- packing_service_requested (Boolean)
- labor_count_requested (Integer)
- created_at (Timestamp)

order_items

- id (UUID, Primary Key)
- order_id (UUID, Foreign Key)
- inventory_item_id (UUID, Foreign Key)
- quantity (Integer)
- is_packed (Boolean)

order_assignments (The Crew)

- id (UUID, Primary Key)
- order_id (UUID, Foreign Key)
- driver_id (UUID, Foreign Key)
- vehicle_id (UUID, Foreign Key)
- laborers (Array of UUIDs) - *Links laborers to specific delivery team.*

- assigned_at (Timestamp)

4. Execution & Support

manifests (Routes)

- id (UUID, Primary Key)
- order_assignment_id (UUID, Foreign Key)
- optimized_route (JSON Path)
- estimated_travel_time (Interval)
- proof_of_delivery_url (String)
- customer_signature_url (String)
- completion_timestamp (Timestamp)

audit_logs

- id (UUID, Primary Key)
- user_id (UUID, Foreign Key)
- action (String)
- details (JSONB)
- timestamp (Timestamp)

customer_feedback

- order_id (UUID, Foreign Key)
- rating (Integer 1-5)
- comment (Text)
- sentiment_score (Float) - *Populated by GenAI.*

5. Financials

transactions

- id (UUID, Primary Key)
- order_id (UUID, Foreign Key)
- amount (Decimal)
- type (Enum: PREPAYMENT, COD, PAYROLL, EXPENSE)
- status (Enum: PENDING, RECONCILED, DISPUTED)
- evidence_url (String) - *OCR Receipts for fuel.*

6. Formal DBML Definition

```
////////////////////////////////////////////////////
// ENUMS
////////////////////////////////////////////////////

Enum user_role {
```

```
  MANAGER
  WAREHOUSE_MANAGER
  DISPATCHER
  DRIVER
  LABORER
  CUSTOMER
}

Enum inventory_category {
  PRODUCT
  UTENSIL
  PACKING_MATERIAL
}

Enum vehicle_type {
  VAN
  SMALL_TRUCK
  LARGE_TRUCK
}

Enum vehicle_status {
  MOVING
  IDLE
  MAINTENANCE
}

Enum order_type {
  COMMERCIAL
  PERSONAL_SHIFT
}

Enum order_status {
  PENDING
  APPROVED
  PICKING
  READY
  DISPATCHED
  COMPLETED
  CANCELLED
}

Enum transaction_type {
  PREPAYMENT
```

```
  COD
  PAYROLL
  EXPENSE
}

Enum transaction_status {
  PENDING
  RECONCILED
  DISPUTED
}

//////////////////////////////////////////////////////
// CORE ORGANIZATIONAL TABLES
//////////////////////////////////////////////////////

Table warehouses {
  id uuid [pk]
  name varchar
  location geography  // Point (Lat, Long)
  address text
  operating_hours json
  capacity_limit int
  created_at timestamp
}

Table users {
  id uuid [pk]
  warehouse_id uuid [ref: > warehouses.id]
  role user_role
  full_name varchar
  email varchar [unique]
  password_hash varchar
  phone_number varchar
  is_active boolean
}

//////////////////////////////////////////////////////
// INVENTORY & RESOURCES
//////////////////////////////////////////////////////

Table inventory_items {
  id uuid [pk]
  warehouse_id uuid [ref: > warehouses.id]
```

```
  sku_code varchar [unique]
  name varchar
  dimensions json   // {l, w, h, weight}
  quantity_on_hand int
  safety_threshold int
  location_code varchar
  category inventory_category
}

Table vehicles {
  id uuid [pk]
  warehouse_id uuid [ref: > warehouses.id]
  license_plate varchar
  type vehicle_type
  seat_capacity int
  load_capacity_kg float
  status vehicle_status
  last_gps_lat float
  last_gps_long float
}

////////////////////////////////////////////////////////
// ORDER & WORKFLOW MANAGEMENT
////////////////////////////////////////////////////////

Table orders {
  id uuid [pk]
  requester_id uuid [ref: > users.id]
  warehouse_id uuid [ref: > warehouses.id]
  order_type order_type
  status order_status
  pickup_address text
  destination_address text
  geofence_polygon json
  total_cost decimal
  packing_service_requested boolean
  labor_count_requested int
  created_at timestamp
}

Table order_items {
  id uuid [pk]
  order_id uuid [ref: > orders.id]
```

```
  inventory_item_id uuid [ref: > inventory_items.id]
  quantity int
  is_packed boolean
}

Table order_assignments {
  id uuid [pk]
  order_id uuid [ref: > orders.id]
  driver_id uuid [ref: > users.id]
  vehicle_id uuid [ref: > vehicles.id]
  laborers uuid[]   // Array of UUIDs
  assigned_at timestamp
}

////////////////////////////////////////////////////////
// EXECUTION & SUPPORT
////////////////////////////////////////////////////////

Table manifests {
  id uuid [pk]
  order_assignment_id uuid [ref: > order_assignments.id]
  optimized_route json
  estimated_travel_time interval
  proof_of_delivery_url varchar
  customer_signature_url varchar
  completion_timestamp timestamp
}

Table audit_logs {
  id uuid [pk]
  user_id uuid [ref: > users.id]
  action varchar
  details jsonb
  timestamp timestamp
}

Table customer_feedback {
  order_id uuid [pk, ref: > orders.id]
  rating int  // 1-5
  comment text
  sentiment_score float
}
```

```
////////////////////////////////////////////////////
// FINANCIALS
////////////////////////////////////////////////////

Table transactions {
  id uuid [pk]
  order_id uuid [ref: > orders.id]
  amount decimal
  type transaction_type
  status transaction_status
  evidence_url varchar
}
```